ÜSLÜ İFADELER

- $-\left[\left(-\frac{1}{2}\right)^{-1}\right]^4 = -(-2)^4 = -2^4 = -16$

- $-4^5 + 3.4^5 - 7.3^4 - 33^4 - 4^5 = 4^5(-1+3-1) + 3^4(-7-3) = 4^5 - 10.3^4$

- $\frac{5^{80}+5^{85}+5^{90}}{5^{85}+5^{80}+5^{75}} = x \quad x = \frac{5^{80}(1+5^5+5^{10})}{5^{75}(1+5^5+5^{10})} = 5^{80-75} = 5^5 \quad \frac{5^5}{5^3} = 5^2 = 25$

- $\frac{14^n+6^n+2^n}{7^n+3^n+1} = \frac{7^n.2^n+2^n.3^n+2^n}{7^n+3^n+1} = \frac{2^n(7^n+3^n+1)}{7^n+3^n+1} = 2^n$

- $(0{,}027)^{5/3}.10^5 \;,\; (3^3)^{5/3}.(10^{-3})^{5/3}.10^5 = 3^5.10^{-5}.10^5 = 3^5$

- $\frac{-9^{-2}.25^{-1}}{125^{-1}.27^{-1}} = \frac{-(3^2)^{-2}.(5^2)^{-1}}{(5^3)^{-1}.(3^3)^{-1}} = \frac{-3^{-4}.5^{-2}}{5^{-3}.3^{-3}} = -\frac{5}{3}$

- $\frac{x^{a+1}}{x^{a-2}}.\left(\frac{x^{-a+2}.x^{2a+3}}{x}\right)^{-1} = x^{a-a+1+2}.(x^{-a+2+2a+3-1})^{-1} = x^3.(x^{a+4})^{-1}$
$= x^3.x^{-a-4} = x^{-a-1}$

- $\left(\frac{a^x}{a^y}\right)^{x-y}.\left(\frac{a^y}{a^x}\right)^{x-y} = \left(\frac{a^x}{a^y}.\frac{a^y}{a^x}\right)^{x-y} = 1^{x-y}$

- $3^x = 2 \;,\; 27^{x+1} - 81^x = ?$

$3^{3x+3} - 3^{4x} = (3^x)^3.3^3 - (3^x)^4 = 8.27 - 16 = 200$

MUTLAK DEĞER

- $-2<x<4$, $|x+2|+|x-4|$

$x+2>0$, $x-4<0 \Rightarrow x+2-x+4=6$

- $\frac{-|2x-y|-|2y-x|}{x+y} = \frac{-2x+y-2y+x}{x+y} = \frac{-x-y}{x+y} = -1$

KÖKLÜ İFADELER

- $\sqrt{0,64}+\sqrt{0,04}-\sqrt[3]{0,027} = \sqrt{\frac{64}{100}}+\sqrt{\frac{4}{100}}-\sqrt[3]{\frac{27}{1000}} = \frac{8}{10}+\frac{2}{10}-\frac{3}{10}$

$=7/10$

- $\frac{\sqrt{75}-\sqrt{27}+\sqrt{12}}{\sqrt{108}+\sqrt{48}} = \frac{\sqrt{3.5^2}-\sqrt{3^3}+\sqrt{2^2.3}}{\sqrt{6^2.3}+\sqrt{4^2.3}} = \frac{5\sqrt{3}-3\sqrt{3}+2\sqrt{3}}{6\sqrt{3}+4\sqrt{3}}$

$= \frac{2\sqrt{3}}{10\sqrt{3}} = 2/10$

- $\sqrt{2}(\sqrt{18}+\sqrt{50}-\sqrt{32}+\sqrt{8})$

$\sqrt{36}+\sqrt{100}-\sqrt{64}+\sqrt{16} = 6+10-8+4$

$=12$

- $\frac{\sqrt[5]{16}}{\sqrt[5]{\frac{1}{2}}} = \sqrt[5]{32} = \sqrt[5]{2^5} = 2$

- $\sqrt{25-25x^2}-\sqrt{64-64x^2}$

$=5\sqrt{1-x^2}-8\sqrt{1-x^2} = -3\sqrt{1-x^2}$

- $\sqrt[5]{\sqrt{\sqrt{\sqrt[3]{\sqrt{2}}}}} = \sqrt[5.2.2.3.2]{2} = \sqrt[120]{2}$

- $\sqrt[3]{2\sqrt[3]{4}} = \sqrt[3]{\sqrt[3]{2^2 3}\,2^2} = \sqrt[16]{2^8}$

$= \sqrt[9]{2^4} = 2^{4/9}$

- $\left(\sqrt[5]{a^2}\right)^4 = \sqrt[5]{a^8} = a^{8/5}$

- $\sqrt{(-4)^2}-\sqrt[3]{(-2)^3}+\sqrt[4]{(-2)^4}+\sqrt[5]{3^5} = 4+2+2+3 = 11$

Globally Yours

- $\sqrt{a^2}+\sqrt{(1-a)^2}-\sqrt{4a^2} = |a|$

$|a|=-a$

- $\sqrt{(b-a)^2}-\sqrt{(2a-b)^2} = |$

$a>0$, $b<0$

- $\frac{2}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$

$|\sqrt{6}|$

- $A=\sqrt{5}-\sqrt{3}$ $\quad \frac{A}{B}+\frac{B}{A}$

$B=\sqrt{5}+\sqrt{3}$

$\frac{(\sqrt{5}-\sqrt{3})^2+(\sqrt{5}+\sqrt{3})^2}{2}+2$

- $3x-5=0 \quad 3x=5$

- $3x+12+x-6 = 10-3x+$

- $\frac{1}{2x}+\frac{3}{x-5}=0$

- $\begin{matrix} -2/ & 2x-y=-1 \\ - & x-2y=4 \\ \hline \pm & \end{matrix} \Rightarrow$ $-4x$ x $-$ $x=$ $y=$

- $\left.\begin{matrix} a+\frac{1}{b}=4 \\ b+\frac{1}{a}=3 \end{matrix}\right\} \frac{a}{b}$ $\quad ab+1=4b$ $\quad ab+1=3a$

$4b=3a$

$\frac{a}{b}=\frac{4}{3}$

$-\sqrt{a^2}+\sqrt{(-a)^2}-\sqrt{4a^2}=|a|+|-a|-|2a|=-a-a+2a=0$

$|a|=-a$

$-\sqrt{(b-a)^2}-\sqrt{(2a-b)^2}=|b-a|-|2a-b|=-b+a-2a+b=-a$

$a>0,\ b<0$

$-\frac{2}{\sqrt{6}}=\frac{2\sqrt{6}}{6}=\frac{\sqrt{6}}{3}$ $\quad$ $-\frac{2\sqrt{3}}{\sqrt{5}-\sqrt{3}}=\frac{2\sqrt{3}(\sqrt{5}+\sqrt{3})}{\sqrt{5}^2-\sqrt{3}^2}=\frac{2\sqrt{15}+6}{2}=\sqrt{15}+3$

$-A=\sqrt{5}-\sqrt{3}$ $\quad B=\sqrt{5}+\sqrt{3}$ $\quad \frac{A}{B}+\frac{B}{A}+2=?$ $\quad \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}+\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}+2$

$\frac{(\sqrt{5}-\sqrt{3})^2+(\sqrt{5}+\sqrt{3})^2}{2}+2=\frac{5-2\sqrt{15}+3+5+2\sqrt{15}+3}{2}+2=8+2=10$

DENKLEMLER

$-3x-5=0$ $\quad 3x=5$ $\quad x=5/3$

$-3x+12+x-6=10-3x+8$ $\quad 7x=14$ $\quad x=2$

$-\frac{1}{2x}+\frac{3}{x-5}=0$ $\quad \frac{1}{2x}=\frac{3}{x-5}$ $\quad -6x=x-5$ $\quad 5=7x$ $\quad x=\frac{5}{7}$

$-\begin{matrix}2x-y=-1 \\ x-2y=4\end{matrix}$ $\Rightarrow$ $\begin{matrix}-4x+2y=2 \\ x-2y=4\end{matrix}$

$-3x=6$

$x=-2$

$y=-3$

$-\begin{matrix}x+6y=162 \\ \frac{y}{x}=\frac{1}{3}\end{matrix}$ $\quad GK=?$

$3y=x$ $\quad 3y+6y=162$

$9y=162$

$y=18$

$-\begin{matrix}a+\frac{1}{b}=4 \\ b+\frac{1}{a}=3\end{matrix}\Big\}\frac{a}{b}$ $\quad ab+1=4b$ $\quad ab+1=3a$

$4b=3a$

$\frac{a}{b}=\frac{4}{3}$

$\begin{matrix}5x+2y=1 \\ -2x+y=1 \\ ax+3y=5\end{matrix}$

$5x+2y=1$

$-4x-2y=-2$

$x=-1$ $\quad y=3$

$-a+9=5$

$-a=-4$

$a=4$